

# Predicting Hospital Readmission or Discharge for Diabetes Patients Using Machine Learning

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## *Final Project Proposal - Datasci 207 (Section 7)*

### Team Info

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### Motivation

Hospital readmissions are a significant challenge for healthcare systems because they increase costs, strain hospital resources, and may indicate gaps in patient care. Patients with diabetes are particularly vulnerable to complications that can lead to repeated hospital visits. Accurately identifying patients at high risk of readmission could help healthcare providers allocate resources more effectively, improve patient outcomes, and reduce unnecessary healthcare expenditures. **This project aims to develop machine learning models that predict whether a diabetic patient will be readmitted or not.** In addition to building accurate predictive models, the project will **investigate which patient characteristics and treatment-related factors are most strongly associated with the risk of readmission.** Understanding these factors can provide valuable insights into potential interventions that may reduce preventable readmissions.

### Data

The project will use the Diabetes 130-US Hospitals dataset from the UCI Machine Learning Repository. The dataset contains records from 101,766 hospital encounters involving diabetic patients across 130 U.S. hospitals between 1999 and 2008. It includes 47 features describing patient demographics, diagnoses, laboratory procedures, medications, hospital stay information, and readmission outcomes.

The target variable indicates whether a patient was readmitted within 30 days, after 30 days, or not readmitted. For this project, the problem will be framed as a binary classification task by predicting whether a patient was readmitted or not. The dataset is large enough to support robust model training and evaluation while also presenting realistic challenges such as missing values, class imbalance, and high-cardinality categorical variables.

**Dataset:** <https://archive.ics.uci.edu/dataset/296/diabetes+130-us+hospitals+for+years+1999-2008>

### Related Work

Recent research has demonstrated the growing importance of machine learning in diabetes prediction and management. A systematic literature review by [Khokhar et al. \(2025\)](#) analyzed 53 studies that

applied machine learning techniques to diabetes-related prediction problems, including disease diagnosis, risk prediction, diabetic retinopathy detection, and patient outcome forecasting. The review further emphasized that healthcare datasets often present challenges such as missing data, class imbalance, and limited generalizability across patient populations. To address these issues, researchers frequently employ techniques such as cross-validation, feature selection, hyperparameter tuning, and ensemble learning methods. <https://doi.org/10.1016/j.artmed.2025.103132>

[Strack et al. \(2025\)](#) also studied the effectiveness of predicting the hospital readmission rates for diabetes encounters using multivariable logistic regression on a measurement of HbA1c which reflects the blood sugar level. The analysis found that by the time of the study, although hospitals were not measuring HbA1c very often, patients who had been measured showed lower chance of readmission, condition on the admission cause in the first place. The application of the measurement could also hint earlier diabetes management, and doctors were paying more attention to controls and treatments for those patients. <https://doi.org/10.1155/2014/781670>

This project builds upon these findings by applying several of the machine learning approaches identified in the literature to the Diabetes 130-US Hospitals dataset.

## **Github repo**

Link: [https://github.com/wip-0/ds207\\_final\\_project](https://github.com/wip-0/ds207_final_project)